Security Assessment Report: Windows Host Scan with Nmap

Executive Summary

Enabling the host firewall reduced the Windows machine's open ports from 4 to 1, fully blocking the RPC, NetBIOS and SMB services.

The assessment was run in an isolated virtual lab using Nmap 7.95 from a Kali Linux machine. The target was scanned twice: once with the Windows firewall enabled and once with it temporarily disabled. This report documents the results, rates the risks and recommends remediation steps.

Environment and Tools

The work was performed inside an isolated VMware network (192.168.138.0/24) between a scanning machine and a target machine.

Item	Details
Scanning host	Kali Linux (192.168.138.132)
Target host	Windows 10 (192.168.138.1)
Tool	Nmap 7.95
Scan date	October 13, 2026

Commands used:

```
sudo nmap -sn 192.168.138.0/24 -oN discovery.txt
sudo nmap -Pn -sV -O 192.168.138.1 -oN scan_after.txt
sudo nmap -Pn -sV -O 192.168.138.1 -oN scan_before.txt
sudo nmap -Pn --script vuln 192.168.138.1 -oN vuln_scan.txt
```

Note: `scan_after` was run with the firewall enabled. `scan_before` was run after temporarily disabling it, and the firewall was re-enabled once scanning finished.

Host Discovery

The discovery scan found 5 live hosts out of 256 addresses, all of them VMware virtual machines.

Address	MAC address	Likely role
192.168.138.1	00:50:56:C0:00:08	Windows target (VMware adapter)
192.168.138.2	00:50:56:F7:11:9E	VMware NAT gateway
192.168.138.133	00:0C:29:00:27:0E	Another virtual machine
192.168.138.254	00:50:56:EC:F8:08	VMware DHCP server
192.168.138.132	Not shown	Scanning host (Kali)

Roles are inferred from addresses and MAC vendors, and were not confirmed by additional scanning.

Scan Results: Firewall Enabled vs Disabled

With the firewall enabled, one port was open and 999 were filtered. With it disabled, four ports were open and 996 were closed.

Port	Service	Firewall on	Firewall off	Severity	Recommendation
135/tcp	MS RPC	Filtered	Open	Medium	Block from the network; allow trusted hosts only
139/tcp	NetBIOS-SSN	Filtered	Open	Medium	Disable NetBIOS over TCP/IP if unused
445/tcp	SMB	Filtered	Open	High	Restrict access with firewall rules, disable SMBv1, apply security updates
5357/tcp	WSDAPI (HTTPAPI 2.0)	Open	Open	Low	Disable Network Discovery on untrusted networks

A filtered port means the firewall silently drops the request, so an attacker cannot tell whether the port exists. A closed port means the host replied with an explicit reset, which confirms the host is alive and shows its services running behind the firewall. The likely operating system in both scans was Windows 10.

Port 5357 stayed open in both states, meaning a default Windows rule allows it on this network.

Vulnerability Scan

The scan confirmed the host is not affected by one vulnerability (MS10-054), while two results were inconclusive.

Script	Result
smb-vuln-ms10-054	false (not vulnerable)
smb-vuln-ms10-061	Connection error, inconclusive
samba-vuln-cve-2012-1182	Connection error, inconclusive

This scan was run with the firewall disabled. The two inconclusive results do not mean the host is safe from those issues; they mean the scan could not complete the check. Re-running the scan or manually verifying the system's patch level is recommended.

Risks, Recommendations and Conclusion

The highest risk is SMB (port 445), which appears open as soon as the firewall is off and is a common target for network attacks.

Recommendations, in priority order:

1. Keep the firewall enabled on all profiles (Domain, Private and Public).
2. Close unnecessary ports with specific inbound rules following the principle of least privilege.
3. Disable SMBv1 and apply security updates regularly.
4. Disable NetBIOS over TCP/IP and Network Discovery on untrusted networks.
5. Monitor scanning activity and suspicious connections with Wazuh.
6. Re-run the inconclusive vulnerability checks once the connection is stable.

Conclusion: the comparison shows the firewall is the host's first line of defense. Without it, RPC, NetBIOS and SMB were exposed; with it, only one low-severity port remained open.